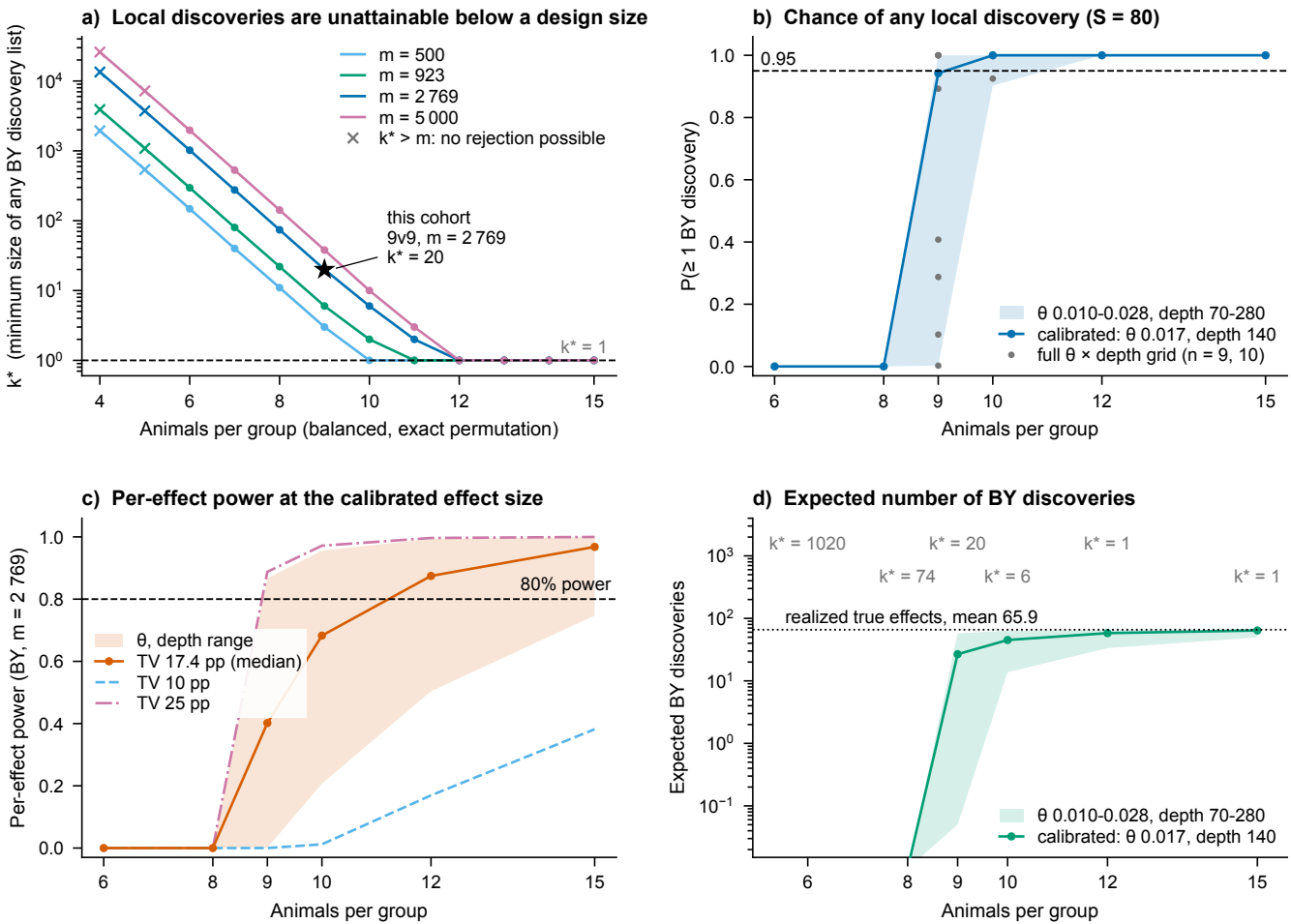
a) Local discoveries are unattainable below a design size
k* (minimum size of any BY discovery list)
Animals per group (balanced, exact permutation)
m = 500
m = 923
m = 2 769
m = 5 000
k* > m: no rejection possible
this cohort
9v9, m = 2 769
k* = 20
k* = 1
b) Chance of any local discovery (S = 80)
P(≥ 1 BY discovery)
0.95
θ 0.010-0.028, depth 70-280
calibrated: θ 0.017, depth 140
full θ × depth grid (n = 9, 10)
Animals per group
c) Per-effect power at the calibrated effect size
Per-effect power (BY, m = 2 769)
80% power
θ, depth range
TV 17.4 pp (median)
TV 10 pp
TV 25 pp
Animals per group
d) Expected number of BY discoveries
Expected BY discoveries
k* = 1020
k* = 74
k* = 20
k* = 6
k* = 1
k* = 1
realized true effects, mean 65.9
θ 0.010-0.028, depth 70-280
calibrated: θ 0.017, depth 140
Animals per group